

RESEARCH ARTICLE

Utilizing rPPG Signal Synchronization and Deep Learning Techniques for Deepfake Video Detection

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ABSTRACT As deepfake technology advances, it poses increasing risks by blurring the line between reality and manipulated content. This challenge particularly threatens the integrity of information in contexts such as propaganda and misinformation. To address this, the paper introduces a novel approach for detecting deepfake videos using physiological signals, specifically remote photoplethysmography (rPPG) signals, obtained from the left and right cheeks. These physiological signals are complex to convincingly alter in tampered video frames, making them reliable indicators of authenticity; thus, we propose the novel method Sync_rPPG, which evaluates signal similarity by employing statistical measures such as Signal to Noise Ratio (SNR), Power Spectral Density (PSD), Mean Absolute Deviation (MAD), Standard Deviation (SD), and Pearson Correlation coefficient (PCC). To enhance signal analysis, we apply the Discrete Wavelet Transform (DWT) to decompose data into time and frequency components, enabling the detection of periodic pattern anomalies. Furthermore, the approach integrates spatial and temporal information, achieving an impressive average accuracy of 94% on CelebDF and 98% on FaceSwap Datasets, thus exceeding the performance of existing state-of-the-art (SOTA) methods. The research conducted makes a significant contribution to the field of media forensics but also opens up new avenues for deep learning applications in biometric analysis and signal processing.

INDEX TERMS Deepfake, photoplethysmography, CNN, LSTM, statistical measures, wavelet transform.

I. INTRODUCTION

Social media provide a valuable platform for expressing ideas and opinions. It facilitates connection and communication with others while fostering community engagement. Additionally, it serves as a form of self-reflection, helping us inward to examine our thoughts, feelings, actions, and motivations. In doing so, it enhances personal awareness and helps individuals better understand themselves and their values. However, it is disconcerting to see the misuse of technology, which undermines the truth and authenticity of the internet. Many celebrities have come across deepfake photos of themselves that are highly unrealistic, and spam accounts have been created with the intent to impersonate and mislead people. On the other hand, its misuse detracts from

the positive aspects of social media. Furthermore, frequent usage has been linked to increased symptoms of depression and anxiety, contributing to a deterioration in student mental health [1]. Thus, while entertainment and social interaction are important, they should never come at the expense of truth and well-being.

In the past, only technical professionals were able to use software like Photoshop and PhotoImpact for designing and morphing, resulting in apparent alterations in media outputs. However, technological advancements and the emergence of numerous online applications have made it possible for individuals without technical skills to create deepfake images and videos [2]. As a result, both celebrities and ordinary individuals may become targets of deepfake content in the future. The increased accessibility of video editing and deepfake creation tools has led to a surge in shared videos, often degrading their quality. Moreover, the

The associate editor coordinating the review of this manuscript and approving it for publication was Antonio J. R. Neves .

rapid growth of technology has intensified concerns about disinformation [3].

Disinformation can negatively impact communities, political leaders and countries, with deepfake technology facilitating the covert insertion of people's faces into inappropriate content [4]. As the creation of fake audio and video becomes more accessible, the risks of assault, extortion, and blackmail are likely to rise. Scammers can exploit realistic digital manipulations to deceive vulnerable individuals, potentially leading to disastrous consequences [5]. The fields of politics and international relations may face the most alarming applications of deepfake technology. However, when applied positively, deepfake technology holds valuable potential. For instance, the medical industry can assist individuals with neurological conditions by implanting a chip in their heads to retrieve and visually display their thoughts on a screen, effectively bringing those thoughts to life [6].

Users frequently share information on platforms such as Facebook, Twitter, and Telegram without verifying its validity, contributing to the rapid spread of content. As information cascades, false information circulates more rapidly than the truth [7]. Deepfake videos that are created using deep learning methods, are known for their capability to process complex and high-dimensional data. Techniques such as Generative Adversarial Networks (GAN) [8] and autoencoders [9], [10] make manipulated images nearly indistinguishable from real ones. Moreover, smartphone applications like AgingBooth, Reflect-Faceswap, Faceapp, and MSQRD (Masquerade), are readily available to the public, making the creation of deepfakes accessible to everyone, which enforces the need to identify and detect promptly.

One promising approach to deepfake detection involves considering that physiological signals, such as heart rate and blood flow, vary significantly among individuals—making them difficult to replicate accurately in synthetic videos. Remote photoplethysmography leverages subtle color changes in facial skin regions, extracted from images or videos, to estimate heart rate. This technique is grounded in the principle that blood flow and oxygenation cause minor variations in skin tone, which are usually imperceptible to the human eye [11], [12] [13], [14]. These variations, particularly noticeable in areas like the cheeks and forehead with rich blood perfusion can be captured using image or video processing methods [15]. The face is often selected because it reliably reflects changes in blood volume. Color information from the selected region is processed to extract a signal corresponding to blood volume fluctuations. Deepfake detection systems employ frequency-domain analysis on the extracted signal, typically from the face region, to estimate heart rate [16]. Although accuracy may be affected by noise sources such as motion artifacts and varying lighting conditions [17], many research had been carried out on signal processing techniques to reduce these effects and enhance the reliability of heart rate estimation. Yet, current deepfake detection methods rarely leverage the uniqueness of individual physiological signals, such as rPPG, especially

under challenging conditions. As a result, distinguishing between real and synthetic content becomes difficult when the visual quality of fake videos is high.

Despite these challenges, rPPG continues to be a reliable method for continuous, non-invasive heart rate monitoring, with practical uses in devices like smartwatches and video-based health applications [18]. With each heartbeat, subtle changes in skin tone most noticeable in the green channel reflect blood volume fluctuations and help reduce the impact of noise. Our approach goes beyond just checking how a video looks. It tracks heartbeat patterns gathered from facial videos over time and points out unusual frequency changes that don't match what a real heartbeat should look like. To ensure the data is stable, we remove slow, gradual shifts in the signal baseline. Together, these steps help us tell if a video is real or artificially made, even when fake videos look almost perfect visually. Although distinguishing real video from fake ones can be challenging, the proposed work offers a potential solution.

The following points outline our main contributions:

- Our approach, Sync_rPPG, introduces a deepfake detection method based on, physiological signals from the left and right cheeks and analyzes metrics like SNR, PSD, MAD, SD, and Pearson correlation to assess signal quality and synchronization, enabling effective deepfake detection.
- We design a hybrid CNN-LSTM architecture integrated with a custom attention mechanism, enabling effective spatial-temporal feature and focusing on subtle physiological inconsistencies.
- Our model achieves 94% accuracy on CelebDF and 98% on FaceSwap. It outperforms existing SOTA methods while maintaining real-time capability with a lightweight design.

The rest of this paper is framed into the following sections. Section II presents an overview of significant related studies in the literature. Section III describes the methodology for reducing noise, measuring heart rate, and extracting rPPG signals. Section IV presents the evaluation results of the proposed model, followed by the conclusion and future research directions.

II. RELATED WORKS

Deepfake video creators use deep learning techniques, specifically GANs [19], [20], to replace one person's face in a video with another. While advances in deep learning have made deepfakes more realistic, they still cannot accurately recreate physiological signals, such as heartbeat rhythms, across various parts of the face. This mismatch is the foundation for exploiting rPPG signals to detect deepfakes. This research analyzes the facial regions within video frames, processed as individual images, to differentiate between authentic and synthetic AI-generated faces, particularly those produced by GANs. Since not all areas of an image are equally important for this task, the proposed method focuses on extracting vital facial features, especially the cheeks, because

they play a crucial role in showing physiological signals like rPPG [21]. Monitoring using Modeling and Bayesian Tracking (MOMBAT) introduces Fourier basis-based pulse modelling to reconstruct pulse signals in low-quality video frames by leveraging signals from high-quality frames. It also integrates a Bayesian tracking framework to correct spurious heart rate estimates, achieving a Pearson correlation of 0.9746 [22].

Using rPPG signals extracted from facial video sequences, [23] addressed motion artifacts between consecutive frames to predict heart rate. An appearance model was used to identify frames containing crucial information for deepfake detection, and a regression layer was employed to estimate the individual's heart rate. Delaunay Triangulation was employed in [24] to map a non-rectangular ROI to a rectangular grid for extracting biological features from the rPPG signal. A support vector machine (SVM) [25] and a CNN were then used as classifiers, achieving accuracies of 94.09% and 91.5% for detecting deepfakes in the FaceForensics++ and Celeb-DF datasets, respectively. DeepRhythm is a fake detector [26] that relies on features extracted using rPPG techniques. While it achieved 100% accuracy on the FF++ database, the method requires a preprocessing stage involving the precise detection of 81 facial landmarks and the use of a color magnification algorithm. However, it obtained only 64% accuracy on the DFDC preview database.

Facial ROIs are often affected by color misalignment, eye movements [27], [28], and variations in appearance such as earrings and facial expressions [29], all of which can degrade the quality of the extracted signal. However, this method requires extracting additional lighting information from the image to ensure that light intensities are sufficient for analyzing eye movements and making subsequent comparisons. The method proposed by [30] extracts information about the leading edges of the eyes, nose, and mouth as facial landmark points for detection using the Regional Deep Facial Network (RDFN). Synthetic images often lack the frequency of saturated (bright) or under-exposed (dark) areas, an observation exploited by [31]. High-pass filtering on the color channels of an image highlights subtle differences [32]. These matrices, derived from high-frequency components, reveal artifacts and patterns differently than natural images.

Deepfake techniques utilize extensive collections of images and videos to train models that can generate photorealistic content. With reels, photographs, and videos of ordinary individuals now widely accessible online, celebrities are no longer the only ones at risk of having their likeness manipulated. As a result, it has become easy for spammers to create fake speeches and spread false information, which is a significant threat to global security [33]. Although edited videos may not be easily identified as altered, small details can provide clues in both visible and technological aspects. The blurred effect caused by editing software doesn't match reality, and examining video reveals whether an edge is blurred naturally or artificially [34]. Real-time

applications like Face-to-Face enable the transfer of a targeted person's facial expressions onto another individual. Similarly, FaceSwap facilitates swapping the faces of two people, regardless of their expressions [35]. The pyVHR (Python-based Virtual Heart Rate) framework offers a robust and comprehensive pipeline for extracting and analyzing heart rate (HR) signals from videos using both traditional and deep learning-based methods. It encompasses all stages of rPPG signal extraction, including skin segmentation, RGB signal processing, blood volume pulse (BVP) estimation, and beats per minute (BPM) calculation. Furthermore, it integrates advanced deep learning models such as MTTTS-CAN, enabling end-to-end HR estimation with high efficiency and accuracy [36].

A two-step approach for analyzing mouth dynamics were introduced in [37]. The first step calculates the mouth aspect ratio (MAR) to locate the mouth region, and the second assesses mouth openness by examining the relative positions of the upper and lower lips. A CNN is employed for object detection, including faces and scenes, in both images and videos. The network performs automated feature extraction through a six-layer architecture. The evaluation uses metrics such as Area Under the Curve (AUC) and Discrete Fourier Transform–Magnitude Feature (DFT-MF), and the experiments involve datasets including Celeb-DF, Deepfake Vid-TIMIT, and UADFV. The DnCNN [38] denoiser, designed for visual noise reduction, identifies distinct noise patterns in deepfake face areas. Using a Siamese architecture, it extracts noise traces from face and background regions, comparing them via a similarity matrix. Due to computational limits, input images resized to 64, with plans to explore varying sizes and videos. Methods like Multi-scale Spatial–Temporal PPG maps combined with Mask-Guided Local Attention and Temporal Transformers effectively capture spatial and temporal inconsistencies in face forgery [39].

In summary, from existing literature, we learned that GAN-generated images often lack the inherent symmetry in authentic videos. Building on this observation, analyzing the symmetry of rPPG signals can effectively detect deepfakes. Human faces consistently produce synchronized physiological signals across both cheeks due to natural blood flow patterns. Conversely, synthetic faces created without accounting for this physiological coherence tend to exhibit temporal or spatial discrepancies in their rPPG signals. From this, we obtained the core innovation of our approach: by examining the asymmetry in rPPG signals between the left and right cheeks, we can reliably identify manipulations. To boost the robustness of our method in the preprocessing stage, we included face detection, normalization, and horizontal flipping to help minimise variations in pose, scale, and rotation and ensure consistency in left-right facial region analysis. This assumption holds particularly well when the cheek regions remain relatively stable, allowing for a meaningful comparison of biological signals that generative models often fail to replicate.

III. PROPOSED WORK

The Sync_rPPG method targets deepfake detection by analyzing physiological irregularities reflected in disrupted rPPG signals across facial regions. Its importance comes from its ability to notice small physical signs, subtle physiological cues, which are notoriously challenging for current deepfake generation algorithms to mimic across many facial regions reliably. The proposed method can work better than regular deepfake detection methods for visual artifacts, giving a stronger and more complete analysis.

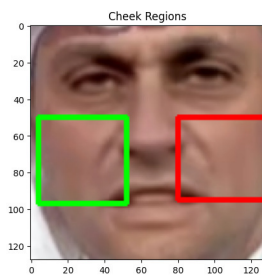


FIGURE 1. Closer view of cheek selection.



FIGURE 2. Wider view of face cropping.

A. DATASETS

For analysis and evaluation, we utilized four benchmark datasets, including FaceForensics++ [40], which comprises 1,000 real YouTube videos and 1,000 synthetically generated videos created using techniques like Face2Face and Faceswap. Celeb-DF v2 [41] dataset includes 590 real and 5,639 synthetic videos of 59 celebrities. The Deepfake Detection Challenge (DFDC) preview dataset, which consists of 1,131 real and fake videos, was used to ensure differences in gender, skin tone, and age. These publicly available benchmark datasets were collected under different acquisition scenarios, including varying lighting conditions in both indoor and outdoor environments, to facilitate deepfake detection research.

B. FACIAL REGION SELECTION

Preprocessing is done by examining video content frame by frame, concentrating on frames where the face is apparent. Accurate rPPG signal extraction requires a clear, unobstructed face, as it depends on recording small color changes caused by heartbeat-induced blood volume fluctuations. We use Dlib's face detection algorithm to detect faces within each frame, which performs reliably under varying lighting conditions, orientations, and facial expressions [42]. Once the face is detected, we apply Dlib's facial landmark predictor, specifically the "shape_predictor_68_face_landmarks.dat"

model, which accurately identifies 68 key facial points, even on previously unseen subjects. Among these landmarks, points 2 to 16 define the outer facial boundary, while those near the nose and under the eyes help outline the top and inner edges of the cheek regions [43], [44]. These points guide us in extracting precise ROIs from the cheeks, which are critical for reliable rPPG signal analysis.

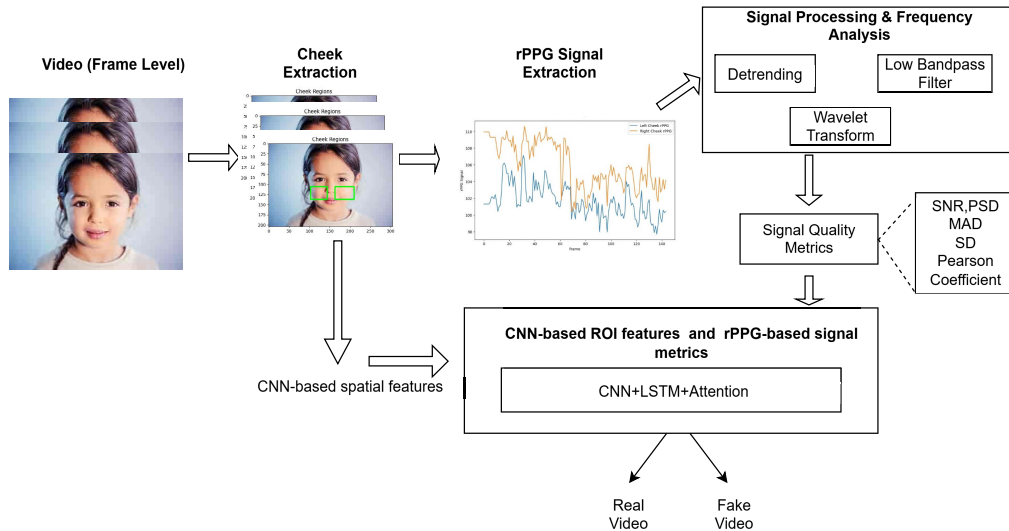
The cheeks were selected for rPPG signal extraction because they offer a valuable combination of high blood perfusion, low mobility, and consistent visibility, as illustrated in Fig 1. The proposed approach is designed to handle input images captured from either close-up or distant views. It automatically identifies the subject by detecting the first face in the frame, then crops the cheek regions accordingly to extract the rPPG signals, as shown in Fig 2. The cheeks are less affected by facial expressions or occlusions such as hair, masks, or glasses than other facial areas. In comparison, other facial regions have certain limitations. While the forehead can provide high rPPG signals, occlusions from hair or shadows reduce its effectiveness. Similarly, the always visible part of the face, nose tends to give weaker signals due to its thinner skin and significantly lower blood flow. Likewise, regions such as the lips and chin are highly dynamic and subject to frequent movement during speech and facial expressions, introducing motion artifacts that degrade the quality of the extracted signals. Full-face approaches increase computational complexity and introduce variability due to differing blood flow patterns across regions. In comparison, selecting cheeks provides a consistent and computationally efficient ROI for capturing physiological information. They also enable asymmetry analysis, as deepfake algorithms frequently fail to synchronize rPPG signals. Thus, cheeks are considered the most suitable area for real-time deepfake identification through rPPG analysis.

We use the green channel of the ROI for rPPG signal processing due to its sensitivity to changes in blood volume [45]. While the ROI may vary across implementations, we typically avoid areas with facial hair, severe shadows, or other obstructions. After identifying the cheeks as ROIs, we process them to generate rPPG signals by averaging the pixel values in the green channel, detrending to remove low-frequency patterns, and applying a band-pass filter to isolate heart rate frequencies. The accuracy of our facial landmark detection directly influences the quality of the rPPG signals we collect. Precise localization allows reliable rPPG estimation even under suboptimal video conditions, thereby enhancing the robustness of our deepfake detection method.

Recognizing the importance of signal quality, this section examines how color channel selection and preprocessing contribute to accurate extraction. The green component is preferred because it effectively captures blood flow dynamics, unlike the Y and V channels, which are more susceptible to distortions particularly in manipulated videos. Table 1 shows that fake videos have higher SNR values in the Y and V channels, which is likely owing to deepfake models improving brightness and color uniformity. The green

TABLE 1. SNR values for different color channels in a single video file.

Video Type	SNR_Green_Left	SNR_Green_Right	SNR_Y_Left	SNR_Y_Right	SNR_V_Left	SNR_V_Right
Fake	27.845	25.312	25.431	23.109	26.984	24.120
Real	29.410	26.784	25.115	22.987	26.227	23.751

**FIGURE 3.** Overall framework.

channel emerges as the most dependable choice for detecting deepfakes using rPPG because it directly captures the subtle variations in blood flow. Unlike the Y and V channels, which are easily susceptible to manipulation because deepfake algorithms often apply global color correction, smoothing, and blending techniques to maintain visual consistency. In fake videos, the green channel remains more robust. The key insight obtained by referring Table. 1 highlights how deepfake algorithms fail to accurately recreate the natural skin color variations seen in real videos, making a strong belief in using the green channel in rPPG-based detection. Since hemoglobin absorption is highest in the green spectrum, real videos consistently show the strongest SNR in this channel, proving it to be the most reliable choice for detecting deepfakes. Another keypoint is noted that deepfake algorithms struggle to recreate genuine heartbeat-induced skin color variations, reinforcing the importance of using the green channel for rPPG-based detection.

We ensure that all frames are enlarged to the exact dimensions (128×128) to offer consistent input to deep learning models, notably CNN, which require fixed-size inputs. Working with smaller, fixed-size frames reduces computing load, speeding up training and inference operations. This approach also limits memory usage, making video frame processing more feasible on hardware with limited resources. This improves our model's learning ability by lowering the chance of vanishing or bursting gradients, which can occur when the input data varies significantly. By standardizing frame size and pixel values, we minimize irrelevant variability, such as differences in lighting or scale,

enabling the model to focus on the relevant features for rPPG signal extraction and deepfake detection. We resize frames to a manageable size to consistently position and scale ROIs like cheeks, simplifying the extraction of meaningful features for analysis. The overall framework is described in detail, outlining each step of the proposed approach, with a visual representation provided in Fig. 3.

C. PHOTOPLETHYSMOGRAPHY SIGNAL EXTRACTION AND ENHANCEMENT

We utilize the inherent symmetry of the human face for our analysis. We hypothesize that in authentic images, the rPPG signals from both cheeks should closely mirror each other, reflecting symmetrical blood flow. Significant discrepancies between these signals within a single frame may indicate manipulation, suggesting the image could be synthetic. Such anomalies are unlikely in genuine human recordings, where physiological responses are naturally symmetrical. To ensure we capture all necessary information, we apply image augmentation techniques to estimate the appearance and rPPG signal of an obscured cheek when only one cheek is visible due to the angle or composition of the frame. We use random rotations of up to ± 20 degrees and horizontal flips with 50% probability. These augmentations allow our model to manage facial orientation and posture differences better. We use Keras' ImageDataGenerator to perform these operations. The augmented frames are processed in batch mode and transformed to uint8 for subsequent analysis, improving our deepfake detection capabilities. Automating ROI identification through facial landmark detection allows

for the examination of large datasets without requiring manual intervention, which is critical for scalable deepfake detection solutions. Research has demonstrated that the green channel has a higher SNR value for detecting blood volume changes than the red or blue channels [16]. This is because green light achieves a balance between being absorbed by hemoglobin and penetrating the skin deeply enough to be reflected back. Signal detrending is an important preprocessing step in rPPG signal analysis. It necessitates eliminating low-frequency patterns from the signal, which can be generated by subject movement, changes in ambient lighting, or other non-physiological variables. Detrending improves signal clarity by removing non-heart rate trends, making the physiological information easier to analyze and comprehend [46].

$$S_t = \frac{1}{N} \sum_{i=1}^N G_t(i) \quad (1)$$

This equation computes the raw rPPG signal S_t by averaging the green channel intensity values over N pixels in the extracted facial region at time t .

Algorithm 1 Preprocessing Algorithm for Sync_rPPG

Input: Video V containing frames
Output: Extracted quality metrics features.
 Initialize frame_count $\leftarrow 0$
while frame_count < max_frames **do**
 $F_t \leftarrow V[t]$
 if F_t is empty **then**
 continue
 end if
 frame_count $\leftarrow$ frame_count + 1
 if frame_count > max_frames **then**
 break
 end if
 Detect face region R using dlib:
 $R \leftarrow \text{dlib}(F_t)$
 Extract left cheek (LC_t) and right cheek (RC_t) via facial landmarks
 Extract green channel:
 $G_t \leftarrow \{LC_t(G), RC_t(G)\}$
 Compute raw rPPG signal:
 $S_t \leftarrow \frac{1}{N} \sum_{i=1}^N G_t(i)$
 Detrend the signal:
 $S'_t \leftarrow S_t - \text{LowPass}(S_t)$
 Apply DWT: Coefficients $\leftarrow \text{wavedec}(S'_t, \text{wavelet})$
 Compute features from coefficients c_m :
 – SNR: Signal-to-noise ratio
 – PSD: Power spectral density (Welch's method)
 – MAD: Median absolute deviation
 – STD: Standard deviation
 – PCC: Pearson correlation (between LC and RC signals)
end while

Cleaner signals allow for more accurate extraction of heart rate and other key metrics, which is critical for applications like health monitoring and deepfake detection. After detrending, the signal is typically blended with other frequency components, including noise. Butterworth bandpass filtering is used to identify frequency components associated with heart rate, which typically range from 0.7 Hz (42 beats per minute) to 4.0 Hz (240 beats per minute). The methodology of focusing on the green channel, combined with signal detrending and Butterworth bandpass filtering, has important implications for both health monitoring and media forensics, including deepfake identification. To the detrended and filtered rPPG signal, Discrete Wavelet Transform (DWT) is used to extract multi-scale frequency components for feature extraction [47], [48], [49]. Though FFT implies periodicity but lacks temporal localization. We decompose the input rPPG signal $s(t)$ using DWT with Daubechies-4 (db4) as the mother wavelet and extract the approximation coefficients that represent the low-frequency components. We compute key statistical features like PSD, STD, MAD, SNR and PCC using the extracted wavelet coefficients. These features serve as inputs for the deepfake detection model. Although the DWT is implemented using the wavedec function from the PyWavelets library in our code, the underlying mathematical formulation is represented as:

$$x(t) = \sum_{m=0}^{M-1} c_m \psi_m(t) \quad (2)$$

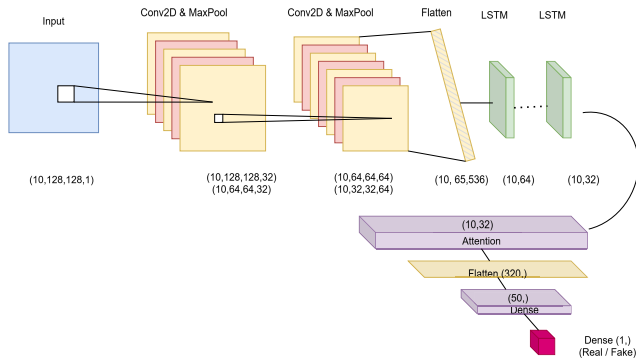
where as:

- $x(t)$ is the original signal,
- $\psi_m(t)$ are the wavelet basis functions — shifted and scaled versions of the mother wavelet,
- c_m are the corresponding wavelet coefficients indicating the contribution of each basis function,
- m denotes the decomposition level or resolution scale.

This equation illustrates how the signal is reconstructed using a weighted combination of basis functions across different scales, corresponding to the approximation and detail coefficients returned by wavedec. Analyzing physiological signals for consistency and realism provides a fresh approach to identifying altered content. The visual plots show some apparent differences, but a quantitative analysis provides a clearer answer. For the fake video, the very low standard deviation ratio (0.44) strongly suggests that the rPPG signals from the left and right cheeks are dissimilar, supporting the hypothesis that the video is likely a deepfake referring Table. 2. The correlation coefficient of the fake video is 0.08, indicating a very weak similarity between the signals. These metrics collectively indicate that the rPPG signals from the left and right cheeks in the fake video are significantly dissimilar, which is consistent with the characteristics of deepfake content.

TABLE 2. Compare real and its corresponding fake video metrics, highlighting significant signal consistency and quality differences.

Metrics	Real Video	Fake Video
Correlation coefficient	0.72	0.08
Mean Squared Error	22.9	11.96
Root Mean Squared Error	4.79	3.46
Standard Deviation Ratio	0.69	0.44
Maximum Cross-Correlation	1553095.941	1291522.357

**FIGURE 4.** Model architecture of sync_rPPG.

D. CALCULATION OF QUALITY METRICS

In our work, we assessed the quality of physiological signals using metrics such as SNR, PSD, MAD, STD, and PCC to differentiate between real and fake videos effectively. For signal processing tasks, including PSD and the PCC, we used the SciPy library, while we employed NumPy to perform statistical computations such as STD and MAD.

- **SNR:** Measures the strength of the rPPG signal compared to the background noise, where a higher SNR indicates better signal quality.

$$\text{SNR} = 10 \cdot \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (3)$$

where as,

- P_{signal} - Power at the dominant frequency (typically corresponding to the heart rate).
- P_{noise} - Average power in surrounding frequencies, excluding a small range around the dominant peak.
- **PSD:** Provides information on the power distribution as a function of frequency as long as the signal has a detectable power level. Here we computed PSD using Welch's method, which employs averaging and smoothing to analyze the underlying random phenomenon. In fraudulent videos, the rPPG signal lack identifiable physiological frequency components, resulting in unusual PSD properties. Understanding frequency distribution is critical for creating bandpass filters that preserve the physiological signal while eliminating undesired noise.

$$\text{PSD} = |\mathcal{F}\{x(t)\}|^2 \quad (4)$$

where as:

- $\mathcal{F}$ denotes the Fourier Transform.
- $x(t)$ represents the time-domain rPPG signal.
- **MAD:** Calculates the average absolute deviation of the rPPG signal from its mean, which captures signal variation. Because it effectively finds outliers and evaluates variability in the rPPG signal, higher readings may reflect fluctuations caused by motion artifacts.

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N |S_i - \bar{S}| \quad (5)$$

where as,

- S_i is the rPPG signal value at time i .
- $\bar{S}$ is the mean of the rPPG signal.
- N is the total number of signal samples.
- **SD:** Quantifies the dispersion of rPPG values around the mean, helping to assess signal consistency and understand the consistency of the extracted signal. A lower SD implies a more stable rPPG signal, which is desirable for accurate physiological measurement.

$$\text{SD} = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_i - \bar{S})^2} \quad (6)$$

- **PCC:** Measures the linear connection between rPPG signals collected from left and right cheeks. Pearson Correlation detects motion-induced disruptions and disregards unreliable signals. By including PCC into our unique approach Sync_rPPG, we can measure the consistency of physiological data across symmetric facial regions and detect motion-induced differences. Fake videos frequently show low rPPG synchronization, whereas actual videos have highly linked signals. Among quality metrics, PCC aid in distinguishing between authentic and false videos by detecting irregularities produced by motion artifacts. The approach enhances classification accuracy and robustness to motion-related distortions by adding them with spatial and temporal information.

$$\text{PCC} = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N (Y_i - \bar{Y})^2}} \quad (7)$$

where as,

- X_i and Y_i are rPPG signal values from two facial regions (left and right cheeks).
- $\bar{X}$ and $\bar{Y}$ are the mean values of the respective signals.
- N is the total number of samples.

E. MODEL DESIGN AND IMPLEMENTATION

The proposed model integrates CNNs, LSTMs, and a custom Attention mechanism to classify input sequences as shown in Fig. 4. To ensure temporal dependencies across frames

in a video sequence, a sliding window-based chunking technique was employed. To ensure that the extracted frames are meaningful and contribute to reliable physiological signal estimation, the selection process is based on face visibility. Only frames in which the face is clearly detected and unobstructed are considered. Using a face detection algorithm (dlib), the video is scanned frame-by-frame, and the 50 most suitable frames, typically those with consistent and frontal face appearances—are chosen either uniformly across the video duration or from segments with minimal motion blur and occlusion. This approach ensures signal stability, which is essential for extracting accurate rPPG features and maintaining model reliability during deepfake detection. Specifically, from each input video, a subset of 50 frames is selected to serve as the input sequence for analysis, overlapping frame segments were extracted using a window size of 10 frames with a stride of 5, resulting in multiple chunks that capture localized temporal variations. We processed each chunk to extract spatial features at the frame level using a TimeDistributed CNN. Then, the LSTM layer learns temporal patterns across the 10-frame chunks to capture fine-grained time dependent cues while preserving frame-wise spatial characteristics. The use of sliding windows ensures broader coverage across the video and enhances the model’s robustness in detecting subtle temporal inconsistencies typical of deepfake videos. The CNN branch extracts spatial features from the input data through two convolutional layers (Conv2D) that process the features with increasing filters (32, 64) to capture local patterns. Each convolutional layer uses max-pooling layers to reduce spatial dimensions and focus on salient features, while dropout layers (0.25) prevent overfitting. The output of the CNN branch is flattened and passed to two stacked LSTM layers to capture temporal dependencies across timesteps. The first LSTM layer has 64 units and returns sequences, while the second LSTM layer has 32 units, further refining temporal patterns. Dropout (0.3) and recurrent dropout (0.3) are applied to enhance regularization. The custom attention mechanism dynamically weighs the importance of features at each timestep by combining features from the LSTM layers with quality metrics. The model dynamically scales the importance of features across timesteps in the attention layer by learning trainable weights W and b , which are used to compute attention scores. These scores are normalized using the sigmoid method, which emphasizes the most essential features. The attention mechanism aggregates the weighted features, highlighting reliable timesteps while suppressing noise and irrelevant inputs. The attended features are passed to fully connected layers for classification. A dense layer with 50 units and ReLU activation refine the features, followed by dropout regularization (0.5) to prevent overfitting. Finally, a dense layer with sigmoid activation generates a binary classification (real or fake). As shown in Table. 3, the model is trained using the hyperparameters listed below.

TABLE 3. Hyperparameter settings.

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Number of Epochs	70
Batch Size	32
CNN Kernel Size	(2, 2)
CNN Activation	ReLU
Final Activation	Sigmoid
Validation Split	20%
Evaluation Metrics	Precision, Accuracy, F1 Score, Recall

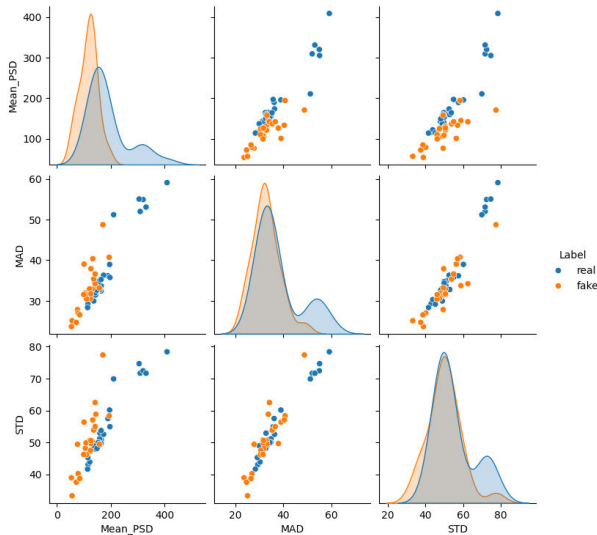


FIGURE 5. The pair plot visualizes the distribution and relationships between different features (Mean of PSD, MAD, STD) for real and fake videos.

IV. RESULT ANALYSIS

A. IMPORTANCE OF QUALITY METRICS

The deep learning model was tested on video datasets to assess whether a video was authentic or not. Our proposed Sync_rPPG framework integrates temporal coherence and frequency based analysis of physiological signals, enabling accurate classification of real and fake videos. In Fig. 5, the Mean_PSD plot indicates the partial separation between real and fake samples. The MAD and STD distributions also show varying degrees of overlap, with fake samples tending to have higher values and a wider distribution. Scatter plots reveal positive correlations between the features but still show significant overlap among real and fake samples. We observe a more precise separation after applying band-pass filtering and detrending as refer to Fig. 6. Actual samples consistently exhibit lower Mean_PSD values, while fake samples show higher ones. The MAD and STD distributions grow increasingly distinct, with false samples showing greater values.

Scatter plots emphasise this distinction, with actual samples forming compact clusters and fraudulent samples seeming more dispersed, especially in the Mean_PSD vs. MAD and Mean_PSD vs. STD plots.

Our Sync_rPPG model introduces a novel, lightweight, and highly effective synchronization-based approach by isolating cheek-based rPPG signals and analyzing their synchronization rather than relying solely on whole-frame quality metrics. Correlation analysis before and after detrend, as shown in Fig. 7 and Fig. 8, reveals deepfake inconsistencies more effectively. It provides physiological reasoning for deepfake classification by capturing coherence in natural rPPG signals.

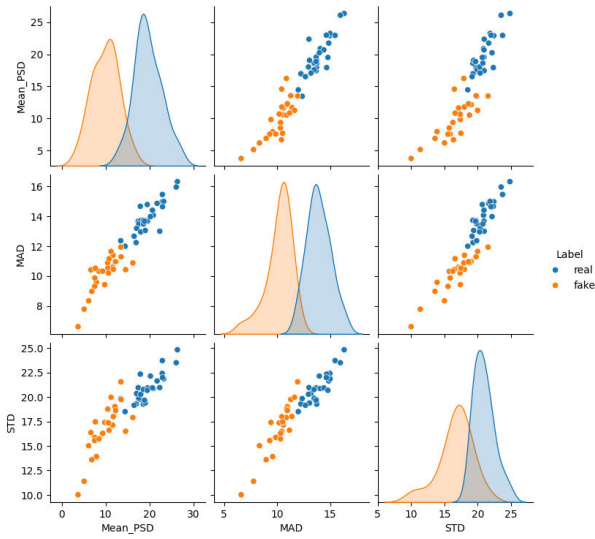


FIGURE 6. The pair plot visualizes the distribution and relationships between different features (Mean of PSD, MAD, STD) for real and fake videos after applying detrend.

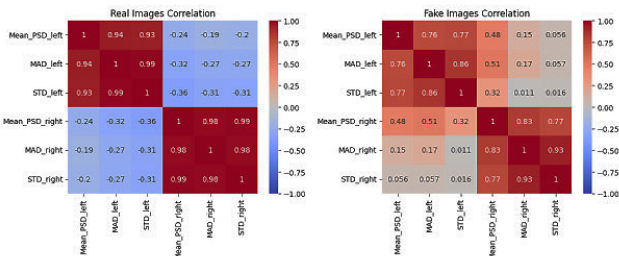


FIGURE 7. Correlation analysis for quality metrics of left and right cheeks for raw videos at frame level.

Fake samples tend to exhibit higher Mean_PSD values, which is a key distinguishing factor, as illustrated in Fig. 8. Also, fake samples exhibit higher MAD values, indicating more significant variability in absolute deviations due to weaker physiological coherence, and higher STD values, indicating irregular signal fluctuations. Unlike conventional CNN or CNN combined with LSTM models, which focus solely on full-frame spatial or temporal patterns, our Sync_rPPG approach integrates signal synchronization

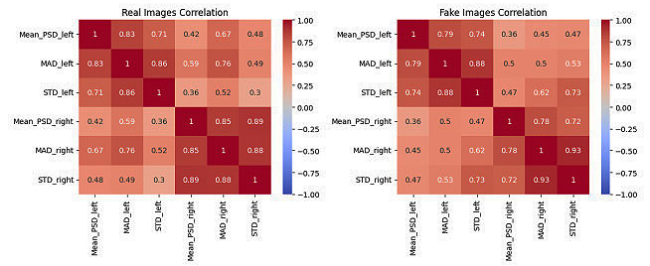


FIGURE 8. Correlation analysis for quality metrics of left and right cheeks for videos at frame level after undergoing detrending.

with spatial and temporal features, enhancing accuracy and efficiency in deepfake detection. By leveraging physiological signal coherence, our method consistently outperforms traditional detection techniques.

B. EVALUATION METRICS

1) ACCURACY

Accuracy gives the value of the model how close it has predicted to the target or the actual value or in other term it can be understood as how many times the model has correctly predicted among all the predictions made. The equation represents the formula used to calculate accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

where as,

TP : True Positive

TN : True Negative

FP : False Positive

FN : False Negative

2) PRECISION

Precision indicates the consistency of results in relation to the model's ability to match the true value with the target label.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

3) RECALL

Recall measures the completeness of positive predictions.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

4) F1 SCORE

The F1 score represents the harmonic average of precision and recall. It provides a single metric that balances both the system of precision and recall.

$$F1 \text{ Score} = \left(\frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} \right) \quad (11)$$

The effectiveness of the model was evaluated using 4-fold cross-validation. The dataset was split into four parts for each round: three parts were combined to form training subsets for model evaluation, while the remaining part was used for validation. Four iterations of this holdout strategy were conducted, and performance metrics were measured

for each round. Table. 4 displays the result of computing across different folds. The evaluation is based on a binary classification system.

TABLE 4. Performance metrics across different folds for FaceSwap dataset in ratio form ranging from (0 - 1).

Measures	Fold 1	Fold 2	Fold 3	Fold 4
Accuracy	0.96	0.96	0.97	0.98
Recall	0.96	0.95	0.98	0.96
Precision	0.98	0.97	0.98	0.97
Specificity	0.95	0.96	0.95	0.94
F1 Score	0.97	0.96	0.98	0.97
False Positive Rate	0.04	0.03	0.04	0.05
False Negative Rate	0.03	0.04	0.01	0.01

Fig. 9 shows the confusion matrix findings for each of the four folds. Effectiveness of the proposed model is assessed for distinguishing between “real” and “fake”, with accuracy, precision, recall, and F1-score. Each metric evaluates the model’s performance from a different perspective.



FIGURE 9. Comparison of different folds used to classify real and fake videos. Each subfigure shows the confusion matrix for a specific fold.

C. RESULT DISCUSSION

The proposed Sync_rPPG model outperforms SOTA deep-fake detection methods by combining multi-domain features: it leverages frequency-domain indicators (SNR, PSD, STD, MAD, and PCC), extracts spatial patterns through CNN architectures, models temporal dependencies with LSTM networks, and employs a novel attention mechanism designed to prioritize salient physiological cues from rPPG signals and suppress irrelevant noise. Traditional deepfake detection

systems usually look for visible irregularities or distortions in the spatial domain essentially, how the pixels or frames differ from what should be present. In contrast, this model uniquely analyses rPPG signals by capturing their changes over time and across various frequency ranges. Wavelet-based transforms are employed because they provide a highly detailed time-frequency representation, enabling the model to pinpoint subtle physiological changes that may be too transient or nuanced for simple spatial analysis. By integrating these time-frequency features, the model becomes more sensitive to the underlying biological signals that real videos naturally exhibit, making it more robust in detecting sophisticated forgeries that spatial artifact-based methods might miss.

Experimentation on the FaceSwap dataset reveals high training accuracy of 98% and expedited convergence facilitated by a compact feature vector of 129 elements, ensuring computational efficiency. Due to computational constraints, we decoupled low-level spatial feature extraction from the main deep learning pipeline by employing a lightweight CNN during preprocessing to encode spatial features from raw facial ROIs (left and right cheeks) [50]. These CNN-derived features, which capture spatial patterns potentially indicative of physiological cues, are not learned from rPPG signals directly but rather from individual video frames. In parallel, we extract signal-based quality metrics from green-channel-based rPPG signals. The resulting spatial and temporal descriptors combined to form a compact 129-dimensional hybrid feature vector. This representation serves as input to a computationally efficient CNN-LSTM-Attention model designed for classifying real versus fake video sequences. Notably, the initial CNN encoder used in preprocessing is not part of the end-to-end training process, which enables reduced model complexity and makes the overall architecture suitable for low-resource environments. As shown in Table. 5, Sync_rPPG stands out as the most lightweight model, with the fewest number of trainable parameters and one of the fastest training times among all the compared methods. Despite its simplicity, it outperforms heavier models, such as CapsuleNet and Mesonet, in both complexity and speed. The proposed model is lightweight and trains faster, resulting in a balance of speed and performance. For systems with limited computing power, it becomes an efficient choice. Consistently high training accuracy obtained in Fig. 10, outperforming well-known pretrained models, strong performance reflects the model’s ability to learn from training data effectively. Throughout the training process, it remains stable and reliable, with minimal fluctuations across epochs. While other models tend to converge slowly or inconsistently, Sync_rPPG steadily improves with each iteration. It also reaches high accuracy much earlier in training, showing a fast and confident learning curve. Fig. 11 presents the training loss curves on the Faceswap dataset over 70 epochs. The proposed Sync_rPPG model demonstrates significantly faster convergence than existing models, reducing the training loss substantially

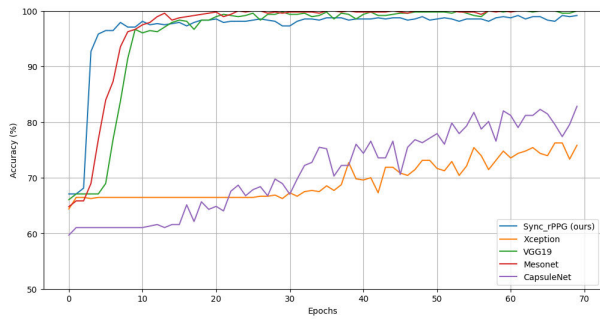


FIGURE 10. The line graph shows the performance of the Faceswap dataset, specifically tracking their training accuracy over 70 epochs.

within the first 10 epochs and maintaining consistently low and stable loss throughout the training process. MesoNet and CapsuleNet didn't perform as well, as their training loss remained higher and showed more fluctuations over time. On the CelebDF dataset, Sync_rPPG achieves an accuracy of 94.17% (as shown in Table. 7) by effectively distinguishing real from fake sequences using the correlation coefficient between left and right cheek rPPG signals. This physiological indicator shows a clear separation, with a mean value of 0.72 for real videos and 0.08 for fake ones. Additionally, the integration of statistical features such as PSD, STD, and PCC improves the model's precision and enhances its robustness against incorrect predictions.

TABLE 5. Comparison of trainable parameters and training time for different models.

Model	Trainable Parameters	Training Time (sec)
Xception	159,458	50.11
VGG19	184,994	36.26
Mesonet	1,538,024	40.01
CapsuleNet	1,118,882	55.27
Synch_rPPG	113,992	34.12

In contrast, existing models, such as DeepRhythm achieved 100% accuracy on the FaceForensics++ dataset but only 64.01% on the Deepfake Detection Challenge (DFDC) dataset, demonstrating limited generalizability as shown in Table. 6. The proposed Sync_rPPG model achieves better performance across all evaluation metrics—accuracy, precision, recall, and F1 score—on the FaceSwap dataset as shown in Fig. 12, outperforming other state-of-the-art models such as Xception, VGG19, MesoNet, and CapsuleNet. Although VGG19 and MesoNet demonstrate relatively strong performance, they slightly underperform in recall and F1 score, generating high-dimensional feature vectors (4096 and 2048 respectively). Where as, Sync_rPPG effectively utilizes only 129 combined features of CNN-based ROI features and statistical features consistently maintain strong performance across multiple datasets, including CelebDF, DFDC, and FaceForensics++, highlighting its adaptability, accuracy, and robustness against evolving deepfake techniques.

TABLE 6. Comparison of physiological-based and non-physiological-based deepfake detection methods.

Method	rPPG-Based	Accuracy (CelebDF)	Accuracy (DFDC)	Detection
Xception [40]	No	75.24%	71.00%	Deepfake
VGG19 [41]	No	67.01%	64.78%	Deepfake
DeepRhythm [26]	Yes	100.00% (FF++)	64.10%	Heart Rate Estimation
FakeCatcher [24]	Yes	94.09%	91.5%	Deepfake
Sync_rPPG (Ours)	Yes	94.17%	90.00%	Deepfake

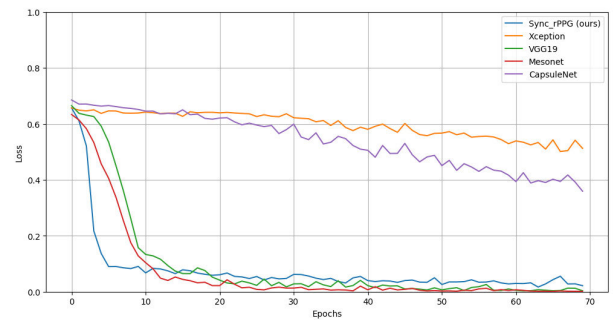


FIGURE 11. Illustrates the training loss for the Faceswap dataset over 70 epochs.

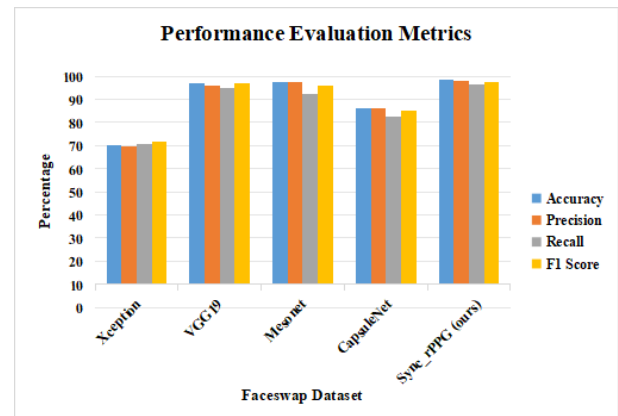


FIGURE 12. Bar chart showing the performance evaluation metrics (Accuracy, Precision, Recall, F1 Score) for different models on FaceSwap dataset.

V. LIMITATIONS AND FUTURE WORK

In future research, we aim to improve rPPG signal extraction for different skin tones, as darker skin reflects less light, making signal capture more difficult. We will also examine how facial features, makeup, and facial hair affect detection accuracy. Occlusions such as glasses, masks, or hands covering the face can obstruct important facial regions, adding further challenges. To address these limitations, we plan to develop adaptive methods that enhance rPPG signal reliability, ensuring accurate deepfake detection across diverse individuals and real-world environments.

TABLE 7. Performance comparison of Deepfake detection models using rPPG signals.

Dataset	Model Implemented with rPPG Signal	Accuracy	Precision	Recall	F1 Score
CelebDF	Xception	75.24	70.85	78.37	74.08
	VGG19	67.01	66.54	69.86	64.53
	Mesonet	87.51	88.93	84.27	86.51
	CapsuleNet	85.52	86.35	87.80	87.09
	Sync_rPPG (Ours)	94.17	95.18	96.34	95.76
DFDC	Xception	88.90	87.19	89.38	86.81
	VGG19	85.01	85.73	90.00	88.80
	Mesonet	84.46	82.00	84.07	88.37
	CapsuleNet	80.96	78.49	79.36	81.54
	Sync_rPPG (Ours)	90.00	89.79	91.23	90.98
FF++ (FaceSwap)	Xception	70.11	69.94	70.56	71.93
	VGG19	97.03	96.00	95.07	97.18
	Mesonet	97.50	97.35	92.29	96.14
	CapsuleNet	85.99	86.00	82.62	85.18
	Sync_rPPG (Ours)	98.33	97.87	96.53	97.69
FF++ (Face2Face)	Xception	52.92	47.05	51.84	49.39
	VGG19	51.26	53.17	49.00	51.17
	Mesonet	48.16	52.55	48.62	50.00
	CapsuleNet	55.58	58.00	52.39	50.29
	Sync_rPPG (Ours)	64.82	62.21	59.97	60.29

VI. CONCLUSION

Detecting deepfakes is a significant challenge in this technological world as fake images and videos spread widely, leading to misinformation and societal hazards. Many existing detection models focus on visual imitations but ignore the difficulty of faking real biological signals. To address this, our proposed model, Sync_rPPG, integrates CNN and LSTM with an attention mechanism, along with physiological features such as STD, PSD, MAD, and Pearson correlation from the rPPG signal. Sync_rPPG presents a robust alternative to traditional deepfake detection methods by analyzing physiological inconsistencies that deepfake generators fail to replicate. Strong correlations between rPPG signals from the left and right cheeks in real videos were demonstrated, whereas deepfake videos exhibit discrepancies. Outperformed or merely similar to the existing models like DeepRhythm and FakeCatcher in accuracy and generalizability. We tried to develop a lightweight and scalable framework suitable for real-time media authentication systems. The results show that Sync_rPPG achieves 94.17% accuracy on CelebDF, 90% on DFDC, and 98% on FaceSwap. Our research highlights the necessity of integrating spatial and temporal physiological signal analysis, which would provide a more reliable and generalizable solution to combat digital media manipulation.

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